Cuaderno de Prácticas

MATEMÁTICAS DISCRETAS

Grado en Ingeniería Informática

CURSO 2025/26

Datos personales

Nombre y apellidos : Francisco Javier Martín - Lunas Escobar
DNI : 26268082
Grupo de teoría : A
Grupo de prácticas : 4

Códigos

In[47]:=

```
(*
Capitulo 7 Producto cartesiano Ejemplo 7.8
*)
CARTESIANO[conjuntos_]:=Module[{i,j,k,cartesianotemp},cartesiano=Table[{conjuntos⟦1,i⟧},{i,Le
Do[cartesianotemp={};
Do[Do[AppendTo[cartesianotemp,Append[cartesiano⟦k⟧,conjuntos⟦i,j⟧]];,{k,Length[cartesiano]}]
cartesiano=cartesianotemp;,{i,2,Length[conjuntos]}];
cartesiano];
(*
Capitulo 8 Relaciones binarias ejemplo 8.1
*)
AnalisisRB[A_,R_]:=Module[{},falloReflexiva={};
Do[If[MemberQ[R,{A⟦n⟧,A⟦n⟧}],Null,AppendTo[falloReflexiva,A⟦n⟧]],{n,Length[A]}];
falloSimetrica={};
Do[If[MemberQ[R,{R⟦m,2⟧,R⟦m,1⟧}],Null,AppendTo[falloSimetrica,{R⟦m,2⟧,R⟦m,1⟧}]],{m,Length[R]
falloTransitiva={};
Do[Do[If[R⟦p,1⟧==R⟦q,2⟧,If[MemberQ[R,{R⟦q,1⟧,R⟦p,2⟧}],Null,AppendTo[falloTransitiva,{{R⟦q,1⟧
falloAntisimetrica={};
Do[If[MemberQ[R,{R⟦r,2⟧,R⟦r,1⟧}]&&!(ToString[R⟦r,1⟧]==ToString[R⟦r,2⟧]),AppendTo[falloAntisi
If[falloReflexiva=={},Print["R es reflexiva"],Print["R no es reflexiva, falla en los elementos
If[falloSimetrica=={},Print["R es simétrica"],Print["R no es simétrica, falla en los pares: ",
If[falloTransitiva=={},Print["R es transitiva"],Print["R no es transitiva, falla en los pares:
If[falloAntisimetrica=={},Print["R es antisimétrica"],Print["R no es antisimétrica, falla en l
If[Union[falloReflexiva,falloSimetrica,falloTransitiva]=={},Print["R es una relación de equiva
If[Union[falloReflexiva,falloAntisimetrica,falloTransitiva]=={},Print["R es una relación de or
(*
```

```
Capitulo 8 Relaciones de equivalencia Ejemplo 8.4
*)
COCIENTE[A_,R_]:=Module[{CONTADORi,CONTADORj,anadir},cociente={{A⟦1⟧}};
 Do[anadir=True;
 Do[If[Intersection[{{A⟦CONTADORi⟧,cociente⟦CONTADORj⟧⟦1⟧}},R]≠{},AppendTo[cociente⟦CONTADORj
  anadir=False;
  Break[];];,{CONTADORj,1,Length[cociente]}];
 If[anadir,AppendTo[cociente,{A⟦CONTADORi⟧}];];,{CONTADORi,2,Length[A]}];
 cociente];
 (**)
COTASSUPERIORES[A_,B_,R_]:=Module[{cotassuperiores,csuper,n,m},cotassuperiores={};
 Do[csuper=True;
 Do[If[Intersection[{{B⟦m⟧,A⟦n⟧}},R]=={},csuper=False;Break[];],{m,1,Length[B]}];
 If[csuper,AppendTo[cotassuperiores,A⟦n⟧]];,{n,1,Length[A]}];
 cotassuperiores]
 (**)
SUPREMO[A_,B_,R_]:=Module[{cotassuperiores,csuper,supremo,mini,n,m},cotassuperiores={};Do[csu
 Do[If[Intersection[{{B⟦m⟧,A⟦n⟧}},R]=={},csuper=False;Break[];],{m,1,Length[B]}];
 If[csuper,AppendTo[cotassuperiores,A⟦n⟧]];,{n,1,Length[A]}];supremo={};Do[mini=True;
 Do[If[MemberQ[R,{cotassuperiores⟦n⟧,cotassuperiores⟦m⟧}], ,mini=False;Break[];],{m,1,Length[
 If[mini,AppendTo[supremo,cotassuperiores⟦n⟧];
 Break[];];,{n,1,Length[cotassuperiores]}];supremo]
 (**)
COTASINFERIORES[A_,B_,R_]:=Module[{cotasinferiores,cinfer,n,m},cotasinferiores={};
 Do[cinfer=True;
 Do[If[Intersection[{{A⟦n⟧,B⟦m⟧}},R]=={},cinfer=False;Break[];],{m,1,Length[B]}];
 If[cinfer,AppendTo[cotasinferiores,A⟦n⟧]];,{n,1,Length[A]}];
 cotasinferiores]
 (**)
INFIMO[A_,B_,R_]:=Module[{cotasinferiores,cinfer,infimo,maxi,n,m},cotasinferiores={};Do[cinfe
 Do[If[Intersection[{{A⟦n⟧,B⟦m⟧}},R]=={},cinfer=False;Break[];],{m,1,Length[B]}];
 If[cinfer,AppendTo[cotasinferiores,A⟦n⟧]];,{n,1,Length[A]}];infimo={};Do[maxi=True;
 Do[If[MemberQ[R,{cotasinferiores⟦m⟧,cotasinferiores⟦n⟧}],,maxi=False;Break[];],{m,1,Length[c
 If[maxi,AppendTo[infimo,cotasinferiores⟦n⟧];Break[];];,{n,1,Length[cotasinferiores]}];
 infimo]
 (**)
MAXIMO[A_,R_]:=Module[{maximo,maxi,n,m},maximo={};
 Do[maxi=True;
 Do[If[MemberQ[R,{A⟦m⟧,A⟦n⟧}],,maxi=False;Break[];],{m,1,Length[A]}];
 If[maxi,AppendTo[maximo,A⟦n⟧];Break[];];,{n,1,Length[A]}];
 maximo]
 (**)
MINIMO[A_,R_]:=Module[{minimo,mini,n,m},minimo={};Do[mini=True;
 Do[
 If[MemberQ[R,{A⟦n⟧,A⟦m⟧}],,mini=False;Break[];],{m,1,Length[A]}];
 If[mini,AppendTo[minimo,A⟦n⟧];Break[];];,{n,1,Length[A]}];minimo]
```

In[5]:=
```
(*
Capitulo 8 Relaciones de orden Ejemplo 8.9
*)
MAXIMALES[A_,R_]:=Module[{maximales,maximal,n,m},maximales={};
 Do[maximal=True;
 Do[If[MemberQ[R,{A⟦n⟧,A⟦m⟧}]&&n≠m,maximal=False;Break[];],{m,1,Length[A]}];
 If[maximal,AppendTo[maximales,A⟦n⟧]];,{n,1,Length[A]}];
 maximales]
```

In[6]:=
```
(*
Capitulo 8 Relaciones de orden Ejemplo 8.9
*)
MINIMALES[A_,R_]:=Module[{minimales,minimal,n,m},minimales={};
 Do[minimal=True;
 Do[If[MemberQ[R,{A⟦m⟧,A⟦n⟧}]&&n≠m,minimal=False;Break[];],{m,1,Length[A]}];
 If[minimal,AppendTo[minimales,A⟦n⟧]];,{n,1,Length[A]}];
 minimales]
```

In[ ]:=
```
(*
Usar para calcular la imagen
Capitulo 7 Aplicaciones Ejemplo 7.16
*)
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A⟦i⟧]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiv
```

Práctica 6

Ejercicio 7.9

Sea A el conjunto formado por todos los dígitos distintos de tu DNI. Calcular:

In[ ]:=
```
A79={2,6,8,0,"N"};
```

a) AxA

In[ ]:=
```
AxA=CARTESIANO[{A79,A79}]
```

Out[ ]=
```
{{2, 2}, {6, 2}, {8, 2}, {0, 2}, {N, 2}, {2, 6}, {6, 6},
 {8, 6}, {0, 6}, {N, 6}, {2, 8}, {6, 8}, {8, 8}, {0, 8}, {N, 8}, {2, 0},
 {6, 0}, {8, 0}, {0, 0}, {N, 0}, {2, N}, {6, N}, {8, N}, {0, N}, {N, N}}
```

b) AxAxA

In[]:=
```
AxAxA=CARTESIANO[{A79,A79,A79}]
```

Out[]=
```
{{2, 2, 2}, {6, 2, 2}, {8, 2, 2}, {0, 2, 2}, {N, 2, 2}, {2, 6, 2}, {6, 6, 2}, {8, 6, 2},
 {0, 6, 2}, {N, 6, 2}, {2, 8, 2}, {6, 8, 2}, {8, 8, 2}, {0, 8, 2}, {N, 8, 2}, {2, 0, 2},
 {6, 0, 2}, {8, 0, 2}, {0, 0, 2}, {N, 0, 2}, {2, N, 2}, {6, N, 2}, {8, N, 2}, {0, N, 2},
 {N, N, 2}, {2, 2, 6}, {6, 2, 6}, {8, 2, 6}, {0, 2, 6}, {N, 2, 6}, {2, 6, 6}, {6, 6, 6},
 {8, 6, 6}, {0, 6, 6}, {N, 6, 6}, {2, 8, 6}, {6, 8, 6}, {8, 8, 6}, {0, 8, 6}, {N, 8, 6},
 {2, 0, 6}, {6, 0, 6}, {8, 0, 6}, {0, 0, 6}, {N, 0, 6}, {2, N, 6}, {6, N, 6}, {8, N, 6},
 {0, N, 6}, {N, N, 6}, {2, 2, 8}, {6, 2, 8}, {8, 2, 8}, {0, 2, 8}, {N, 2, 8}, {2, 6, 8},
 {6, 6, 8}, {8, 6, 8}, {0, 6, 8}, {N, 6, 8}, {2, 8, 8}, {6, 8, 8}, {8, 8, 8}, {0, 8, 8},
 {N, 8, 8}, {2, 0, 8}, {6, 0, 8}, {8, 0, 8}, {0, 0, 8}, {N, 0, 8}, {2, N, 8}, {6, N, 8},
 {8, N, 8}, {0, N, 8}, {N, N, 8}, {2, 2, 0}, {6, 2, 0}, {8, 2, 0}, {0, 2, 0}, {N, 2, 0},
 {2, 6, 0}, {6, 6, 0}, {8, 6, 0}, {0, 6, 0}, {N, 6, 0}, {2, 8, 0}, {6, 8, 0}, {8, 8, 0},
 {0, 8, 0}, {N, 8, 0}, {2, 0, 0}, {6, 0, 0}, {8, 0, 0}, {0, 0, 0}, {N, 0, 0}, {2, N, 0},
 {6, N, 0}, {8, N, 0}, {0, N, 0}, {N, N, 0}, {2, 2, N}, {6, 2, N}, {8, 2, N}, {0, 2, N},
 {N, 2, N}, {2, 6, N}, {6, 6, N}, {8, 6, N}, {0, 6, N}, {N, 6, N}, {2, 8, N},
 {6, 8, N}, {8, 8, N}, {0, 8, N}, {N, 8, N}, {2, 0, N}, {6, 0, N}, {8, 0, N},
 {0, 0, N}, {N, 0, N}, {2, N, N}, {6, N, N}, {8, N, N}, {0, N, N}, {N, N, N}}
```

c) P(A)

In[]:=
```
PartesDeA=Subsets[A79]
```

Out[]=
```
{{}, {2}, {6}, {8}, {0}, {N}, {2, 6}, {2, 8}, {2, 0}, {2, N}, {6, 8}, {6, 0},
 {6, N}, {8, 0}, {8, N}, {0, N}, {2, 6, 8}, {2, 6, 0}, {2, 6, N}, {2, 8, 0},
 {2, 8, N}, {2, 0, N}, {6, 8, 0}, {6, 8, N}, {6, 0, N}, {8, 0, N}, {2, 6, 8, 0},
 {2, 6, 8, N}, {2, 6, 0, N}, {2, 8, 0, N}, {6, 8, 0, N}, {2, 6, 8, 0, N}}
```

d) Una partición de partes de AxA con tres elementos

In[∘]:=
```
A79d={6,2};
X=Subsets[CARTESIANO[{A79d,A79d}]]
A1={};
A2={};
A3={};
For[i=1, i≤Length[X],i++,
    If[Mod[i,3]==0,
        AppendTo[A1,X[[i]]]
        ];
    If[Mod[i,3]==1,
        AppendTo[A2,X[[i]]]
        ];
    If[Mod[i,3]==2,
        AppendTo[A3,X[[i]]]
        ];
]
W={A1,A2,A3}
```

Out[∘]=
```
{{}, {{6, 6}}, {{2, 6}}, {{6, 2}}, {{2, 2}}, {{6, 6}, {2, 6}},
 {{6, 6}, {6, 2}}, {{6, 6}, {2, 2}}, {{2, 6}, {6, 2}}, {{2, 6}, {2, 2}},
 {{6, 2}, {2, 2}}, {{6, 6}, {2, 6}, {6, 2}}, {{6, 6}, {2, 6}, {2, 2}},
 {{6, 6}, {6, 2}, {2, 2}}, {{2, 6}, {6, 2}, {2, 2}}, {{6, 6}, {2, 6}, {6, 2}, {2, 2}}}
```

Out[∘]=
```
{{{{2, 6}}, {{6, 6}, {2, 6}}, {{2, 6}, {6, 2}}, {{6, 6}, {2, 6}, {6, 2}},
   {{2, 6}, {6, 2}, {2, 2}}}, {{}, {{6, 2}}, {{6, 6}, {6, 2}}, {{2, 6}, {2, 2}},
   {{6, 6}, {2, 6}, {2, 2}}, {{6, 6}, {2, 6}, {6, 2}, {2, 2}}},
 {{{6, 6}}, {{2, 2}}, {{6, 6}, {2, 2}}, {{6, 2}, {2, 2}}, {{6, 6}, {6, 2}, {2, 2}}}}
```

Compruebo que ningún elemento de la partición es vacío

In[∘]:=
```
A1=={}
A2=={}
A3=={}
```

Out[∘]=
```
False
```

Out[∘]=
```
False
```

Out[∘]=
```
False
```

Compruebo que son disjuntos 2 a 2

In[∘]:=
```
Intersection[A1,A2]=={}
Intersection[A1,A3]=={}
Intersection[A2,A3]=={}
```

Out[∘]=
```
True
```

Out[∘]=
```
True
```

Out[∘]=
```
True
```

Compruebo que la unión da el total

In[∘]:=
```
SubsetQ[Union[A1,A2,A3],X]&&SubsetQ[X,Union[A1,A2,A3]]
```

Out[∘]=
```
True
```

Ejercicio 7.14

Sea A el conjunto formado por todas las letras distintas que aparecen en tu primer apellido y B el conjunto formado por todos los dígitos distintos de tu DNI, calcular si es posible:

In[∘]:=
```
A714={"M","A","R","T","I","N","L","U","S"};
B714={2,6,8,0,"N"};
```

a) AxB y AxAxB

In[∘]:=
```
CARTESIANO[{A714,B714}]
```

Out[∘]=
```
{{M, 2}, {A, 2}, {R, 2}, {T, 2}, {I, 2}, {N, 2}, {L, 2}, {U, 2}, {S, 2},
 {M, 6}, {A, 6}, {R, 6}, {T, 6}, {I, 6}, {N, 6}, {L, 6}, {U, 6}, {S, 6},
 {M, 8}, {A, 8}, {R, 8}, {T, 8}, {I, 8}, {N, 8}, {L, 8}, {U, 8}, {S, 8},
 {M, 0}, {A, 0}, {R, 0}, {T, 0}, {I, 0}, {N, 0}, {L, 0}, {U, 0}, {S, 0},
 {M, N}, {A, N}, {R, N}, {T, N}, {I, N}, {N, N}, {L, N}, {U, N}, {S, N}}
```

In[∘]:=
```
CARTESIANO[{A714,A714,B714}]
```

Out[∘]=
```
{{M, M, 2}, {A, M, 2}, {R, M, 2}, {T, M, 2}, {I, M, 2}, {N, M, 2}, {L, M, 2}, {U, M, 2},
 {S, M, 2}, {M, A, 2}, {A, A, 2}, {R, A, 2}, {T, A, 2}, {I, A, 2}, {N, A, 2}, {L, A, 2},
 {U, A, 2}, {S, A, 2}, {M, R, 2}, {A, R, 2}, {R, R, 2}, {T, R, 2}, {I, R, 2}, {N, R, 2},
 {L, R, 2}, {U, R, 2}, {S, R, 2}, {M, T, 2}, {A, T, 2}, {R, T, 2}, {T, T, 2}, {I, T, 2},
 {N, T, 2}, {L, T, 2}, {U, T, 2}, {S, T, 2}, {M, I, 2}, {A, I, 2}, {R, I, 2}, {T, I, 2},
 {I, I, 2}, {N, I, 2}, {L, I, 2}, {U, I, 2}, {S, I, 2}, {M, N, 2}, {A, N, 2}, {R, N, 2},
 {T, N, 2}, {I, N, 2}, {N, N, 2}, {L, N, 2}, {U, N, 2}, {S, N, 2}, {M, L, 2}, {A, L, 2},
 {R, L, 2}, {T, L, 2}, {I, L, 2}, {N, L, 2}, {L, L, 2}, {U, L, 2}, {S, L, 2}, {M, U, 2},
 {A, U, 2}, {R, U, 2}, {T, U, 2}, {I, U, 2}, {N, U, 2}, {L, U, 2}, {U, U, 2}, {S, U, 2},
 {M, S, 2}, {A, S, 2}, {R, S, 2}, {T, S, 2}, {I, S, 2}, {N, S, 2}, {L, S, 2}, {U, S, 2},
```

```
{S, S, 2}, {M, M, 6}, {A, M, 6}, {R, M, 6}, {T, M, 6}, {I, M, 6}, {N, M, 6}, {L, M, 6},
{U, M, 6}, {S, M, 6}, {M, A, 6}, {A, A, 6}, {R, A, 6}, {T, A, 6}, {I, A, 6}, {N, A, 6},
{L, A, 6}, {U, A, 6}, {S, A, 6}, {M, R, 6}, {A, R, 6}, {R, R, 6}, {T, R, 6}, {I, R, 6},
{N, R, 6}, {L, R, 6}, {U, R, 6}, {S, R, 6}, {M, T, 6}, {A, T, 6}, {R, T, 6}, {T, T, 6},
{I, T, 6}, {N, T, 6}, {L, T, 6}, {U, T, 6}, {S, T, 6}, {M, I, 6}, {A, I, 6}, {R, I, 6},
{T, I, 6}, {I, I, 6}, {N, I, 6}, {L, I, 6}, {U, I, 6}, {S, I, 6}, {M, N, 6}, {A, N, 6},
{R, N, 6}, {T, N, 6}, {I, N, 6}, {N, N, 6}, {L, N, 6}, {U, N, 6}, {S, N, 6}, {M, L, 6},
{A, L, 6}, {R, L, 6}, {T, L, 6}, {I, L, 6}, {N, L, 6}, {L, L, 6}, {U, L, 6}, {S, L, 6},
{M, U, 6}, {A, U, 6}, {R, U, 6}, {T, U, 6}, {I, U, 6}, {N, U, 6}, {L, U, 6}, {U, U, 6},
{S, U, 6}, {M, S, 6}, {A, S, 6}, {R, S, 6}, {T, S, 6}, {I, S, 6}, {N, S, 6}, {L, S, 6},
{U, S, 6}, {S, S, 6}, {M, M, 8}, {A, M, 8}, {R, M, 8}, {T, M, 8}, {I, M, 8}, {N, M, 8},
{L, M, 8}, {U, M, 8}, {S, M, 8}, {M, A, 8}, {A, A, 8}, {R, A, 8}, {T, A, 8}, {I, A, 8},
{N, A, 8}, {L, A, 8}, {U, A, 8}, {S, A, 8}, {M, R, 8}, {A, R, 8}, {R, R, 8}, {T, R, 8},
{I, R, 8}, {N, R, 8}, {L, R, 8}, {U, R, 8}, {S, R, 8}, {M, T, 8}, {A, T, 8}, {R, T, 8},
{T, T, 8}, {I, T, 8}, {N, T, 8}, {L, T, 8}, {U, T, 8}, {S, T, 8}, {M, I, 8}, {A, I, 8},
{R, I, 8}, {T, I, 8}, {I, I, 8}, {N, I, 8}, {L, I, 8}, {U, I, 8}, {S, I, 8}, {M, N, 8},
{A, N, 8}, {R, N, 8}, {T, N, 8}, {I, N, 8}, {N, N, 8}, {L, N, 8}, {U, N, 8}, {S, N, 8},
{M, L, 8}, {A, L, 8}, {R, L, 8}, {T, L, 8}, {I, L, 8}, {N, L, 8}, {L, L, 8}, {U, L, 8},
{S, L, 8}, {M, U, 8}, {A, U, 8}, {R, U, 8}, {T, U, 8}, {I, U, 8}, {N, U, 8}, {L, U, 8},
{U, U, 8}, {S, U, 8}, {M, S, 8}, {A, S, 8}, {R, S, 8}, {T, S, 8}, {I, S, 8}, {N, S, 8},
{L, S, 8}, {U, S, 8}, {S, S, 8}, {M, M, 0}, {A, M, 0}, {R, M, 0}, {T, M, 0}, {I, M, 0},
{N, M, 0}, {L, M, 0}, {U, M, 0}, {S, M, 0}, {M, A, 0}, {A, A, 0}, {R, A, 0}, {T, A, 0},
{I, A, 0}, {N, A, 0}, {L, A, 0}, {U, A, 0}, {S, A, 0}, {M, R, 0}, {A, R, 0}, {R, R, 0},
{T, R, 0}, {I, R, 0}, {N, R, 0}, {L, R, 0}, {U, R, 0}, {S, R, 0}, {M, T, 0}, {A, T, 0},
{R, T, 0}, {T, T, 0}, {I, T, 0}, {N, T, 0}, {L, T, 0}, {U, T, 0}, {S, T, 0}, {M, I, 0},
{A, I, 0}, {R, I, 0}, {T, I, 0}, {I, I, 0}, {N, I, 0}, {L, I, 0}, {U, I, 0}, {S, I, 0},
{M, N, 0}, {A, N, 0}, {R, N, 0}, {T, N, 0}, {I, N, 0}, {N, N, 0}, {L, N, 0}, {U, N, 0},
{S, N, 0}, {M, L, 0}, {A, L, 0}, {R, L, 0}, {T, L, 0}, {I, L, 0}, {N, L, 0}, {L, L, 0},
{U, L, 0}, {S, L, 0}, {M, U, 0}, {A, U, 0}, {R, U, 0}, {T, U, 0}, {I, U, 0}, {N, U, 0},
{L, U, 0}, {U, U, 0}, {S, U, 0}, {M, S, 0}, {A, S, 0}, {R, S, 0}, {T, S, 0}, {I, S, 0},
{N, S, 0}, {L, S, 0}, {U, S, 0}, {S, S, 0}, {M, M, N}, {A, M, N}, {R, M, N}, {T, M, N},
{I, M, N}, {N, M, N}, {L, M, N}, {U, M, N}, {S, M, N}, {M, A, N}, {A, A, N}, {R, A, N},
{T, A, N}, {I, A, N}, {N, A, N}, {L, A, N}, {U, A, N}, {S, A, N}, {M, R, N}, {A, R, N},
{R, R, N}, {T, R, N}, {I, R, N}, {N, R, N}, {L, R, N}, {U, R, N}, {S, R, N}, {M, T, N},
{A, T, N}, {R, T, N}, {T, T, N}, {I, T, N}, {N, T, N}, {L, T, N}, {U, T, N}, {S, T, N},
{M, I, N}, {A, I, N}, {R, I, N}, {T, I, N}, {I, I, N}, {N, I, N}, {L, I, N}, {U, I, N},
{S, I, N}, {M, N, N}, {A, N, N}, {R, N, N}, {T, N, N}, {I, N, N}, {N, N, N}, {L, N, N},
{U, N, N}, {S, N, N}, {M, L, N}, {A, L, N}, {R, L, N}, {T, L, N}, {I, L, N}, {N, L, N},
{L, L, N}, {U, L, N}, {S, L, N}, {M, U, N}, {A, U, N}, {R, U, N}, {T, U, N},
{I, U, N}, {N, U, N}, {L, U, N}, {U, U, N}, {S, U, N}, {M, S, N}, {A, S, N},
{R, S, N}, {T, S, N}, {I, S, N}, {N, S, N}, {L, S, N}, {U, S, N}, {S, S, N}}
```

b) Una aplicación entre A y P(B) que sea inyectiva y otra entre P(B) y A que sea sobreyectiva.

In[ ]:=
```
Dominio=A714;
Codominio=Subsets[B714];
If[Length[Dominio]≤Length[Codominio], (*Si el dominio tiene menos elementos, puedo definir un
    For[i=1, i≤Length[Dominio], i++,
        f[Dominio⟦i⟧]=Codominio⟦i⟧;
    ]
    ,
    Print["No se puede definir una aplicación inyectiva"];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A⟦i⟧]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiv
```

```
El conjunto imagen es: {{}, {0}, {2}, {6}, {8}, {N}, {2, 0}, {2, 6}, {2, 8}}
No es sobreyectiva
Es inyectiva
No es biyectiva
```

In[ ]:=
```
Clear;
```

In[ ]:=
```
Dominio=Subsets[B714];
Codominio=A714;
If[Length[Dominio]≥Length[Codominio], (*Si el dominio tiene más elementos, puedo definir una
    For[i=1, i≤Length[Dominio], i++,
        f[Dominio⟦i⟧]=Codominio⟦Mod[i,Length[Codominio]]+1⟧; (*ojito índice del codominio cu
    ]
    ,
    Print["No se puede definir una aplicación Sobreyectiva"];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A⟦i⟧]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiv
```

```
El conjunto imagen es: {A, I, L, M, N, R, S, T, U}
Es sobreyectiva
No es inyectiva
No es biyectiva
```

c) Una aplicación no inyectiva de AxB en P(AxB)

In[∘]:=
```
Dominio=CARTESIANO[{A714,B714}];
c=CARTESIANO[{A714,B714}];
Codominio=Subsets[c];
For[i=1, i≤Length[Dominio], i++,
        f[Dominio[[i]]]=Codominio[[1]];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A[[i]]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiv
```

General: The current computation was aborted because there was insufficient memory available to complete the computation.

Throw: Uncaught SystemException returned to top level. Can be caught with Catch[..., _SystemException].

Out[∘]=

```
SystemException[MemoryAllocationFailure]
```

Part: Part specification Codominio[[1]] is longer than depth of object.

```
El conjunto imagen es: {Codominio[[1]]}
No es sobreyectiva
No es inyectiva
No es biyectiva
```

d) Una aplicación no sobreyectiva de P(AxB) en AxB

In[∘]:=
```
Dominio=Subsets[CARTESIANO[{X,Y}]];
Codominio=CARTESIANO[{X,Y}];
For[i=1, i≤Length[Dominio], i++,
        f[Dominio[[i]]]=Codominio[[1]];
]
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A[[i]]]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiv
```

Part: Part 1 of {} does not exist.

```
El conjunto imagen es: {{}[[1]]}
No es sobreyectiva
Es inyectiva
No es biyectiva
```

Ejercicio 18 Rel 2

Sean A={a,b,c}, B={1,2,3}, A1={a,b} y B1={2,3}. Definir, si existe, una aplicación biyectiva f:AxP(B1) -> P(A1)xB y hacer todas las comprobaciones explícitamente

In[∘]:=
```
A18={"a","b","c"};
B18={1,2,3};
A181={"a","b"};
B181={2,3};
Dominio=CARTESIANO[{A18,Subsets[B181]}];
Codominio=CARTESIANO[{Subsets[A181],B18}];
Dominio
Codominio
A=Dominio;
B=Codominio;
Imagen={};
Do[Imagen=Union[Imagen,Append[{},f[A⟦i⟧]]],{i,1,Length[A]}];
Print["El conjunto imagen es: ",Imagen];
If[Length[Imagen]==Length[B],Print["Es sobreyectiva"],Print["No es sobreyectiva"]];
If[Length[A]==Length[Imagen],Print["Es inyectiva"],Print["No es inyectiva"]];
If[Length[A]==Length[B]&&Length[B]==Length[Imagen],Print["Es biyectiva"],Print["No es biyectiv
```

Out[∘]=
```
{{a, {}}, {b, {}}, {c, {}}, {a, {2}}, {b, {2}}, {c, {2}},
 {a, {3}}, {b, {3}}, {c, {3}}, {a, {2, 3}}, {b, {2, 3}}, {c, {2, 3}}}
```

Out[∘]=
```
{{{}, 1}, {{a}, 1}, {{b}, 1}, {{a, b}, 1}, {{}, 2}, {{a}, 2},
 {{b}, 2}, {{a, b}, 2}, {{}, 3}, {{a}, 3}, {{b}, 3}, {{a, b}, 3}}
```

```
El conjunto imagen es: {{{}, 1}, {{}, 2}, {{}, 3}, {{a}, 1}, {{a}, 2},
  {{a}, 3}, {{b}, 1}, {{b}, 2}, {{b}, 3}, {{a, b}, 1}, {{a, b}, 2}, {{a, b}, 3}}
Es sobreyectiva
Es inyectiva
Es biyectiva
```

In[∘]:=
```
If[Length[Dominio]==Length[Codominio], (*Compruebo que dominio y codominio tienen los mismos e
    For[i=1, i≤Length[Dominio], i++,
        f[Dominio⟦i⟧]=Codominio⟦i⟧;
    ]
    ,
    Print["No se puede definir una aplicación biyectiva"];
]
```

Práctica 7

Ejercicio 8.1

Sea A={a,b,c,d,e}, consideramos la relación binaria R={(a,b),(b,c),(d,e)}

In[ ]:=
```
A81={"a","b","c","d","e"};
```

a) Completar la relación binaria R para que sea una relación de orden total

In[ ]:=
```
R81a={
{"a","a"},{"a","b"},{"a","c"},{"a","d"},{"a","e"},
{"b","b"},{"b","c"},{"b","d"},{"b","e"},
{"c","c"},{"c","e"},{"c","d"},
{"d","d"},{"d","e"},
{"e","e"}
};
AnalisisRB[A81,R81a]
```

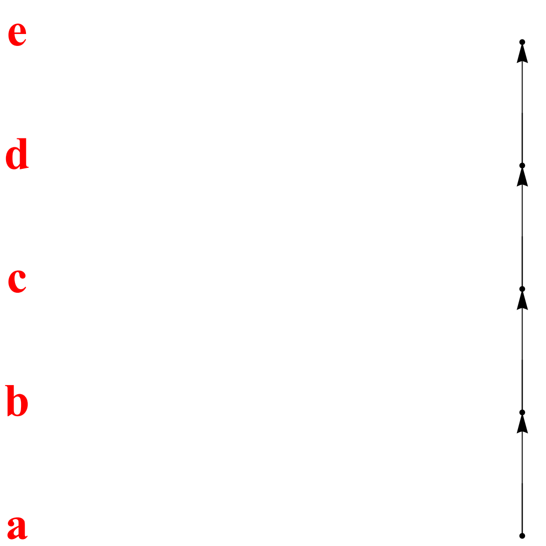
```
R es reflexiva

R no es simétrica, falla en los pares:
 {{b, a}, {c, a}, {d, a}, {e, a}, {c, b}, {d, b}, {e, b}, {e, c}, {d, c}, {e, d}}

R es transitiva

R es antisimétrica

R no es relación de equivalencia

R es una relación de orden
```

In[∘]:=

```
A=A81;
R=R81a;
Clear[Coord];
tabla=Table[0,{i1,Length[A]},{j1,3}];
B=A;
t1=1;
nivel=0;
While[B≠{},minimales={};nivel++;
 Do[minimal=True;
 Do[If[Intersection[{{B⟦m1⟧,B⟦n1⟧}},R]≠{}&&n1≠m1,minimal=False];,{m1,1,Length[B]}];
 If[minimal,AppendTo[minimales,B⟦n1⟧];
 tabla⟦t1⟧={nivel,B⟦n1⟧,0};
 t1++;];,{n1,1,Length[B]}];
 B=Complement[B,minimales];];
R1={};
Do[AppendTo[R1,{A⟦i1⟧,A⟦i1⟧}],{i1,1,Length[A]}];
R=Complement[R,R1];
R1={};
Do[Do[If[Length[Intersection[R,{{R⟦k1,1⟧,A⟦j1⟧},{A⟦j1⟧,R⟦k1,2⟧}}]]==2,R1=Union[R1,{R⟦k1⟧}];]
R=Complement[R,R1];
puntos={};t1=0;
Do[cont=0;
 Do[If[tabla⟦i1,1⟧==j1,cont=cont+1],{i1,1,Length[A]}];
 Do[t1++;
 puntos=Union[puntos,{Text[Style[tabla⟦t1,2⟧,Large,Bold,Red,Background→None,FontFamily→Times]
 tabla⟦t1,3⟧=k1-(cont/2);,{k1,1,cont}],{j1,1,tabla⟦Length[A],1⟧}];
Coord[elem_]:=Do[If[elem==tabla⟦h1,2⟧,Coord[elem]={tabla⟦h1,3⟧,tabla⟦h1,1⟧}],{h1,1,Length[A]
Do[Coord[A⟦i1⟧],{i1,1,Length[A]}];
Do[AppendTo[puntos,Arrow[{Coord[R⟦t1,1⟧],Coord[R⟦t1,2⟧]}]];,{t1,1,Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos],AspectRatio→1,Background→None]
```

Diagrama de orden:

Out[∘]=



Como se observa en el diagrama de Hasse el diagrama de orden es de una relación de orden total

b) Completar la relación binaria R para que sea reflexiva y transitiva, pero no sea simétrica y antisimétrica

In[ ]:=
```
R81b={
{"a","a"},{"a","b"},{"a","c"},{"a","d"},{"a","e"},
{"b","b"},{"b","c"},{"b","d"},{"b","e"},{"b","a"},
{"c","c"},{"c","e"},{"c","d"},
{"d","d"},{"d","e"},
{"e","e"}
};
AnalisisRB[A81,R81b]
```

```
R es reflexiva

R no es simétrica, falla en los pares:
 {{c, a}, {d, a}, {e, a}, {c, b}, {d, b}, {e, b}, {e, c}, {d, c}, {e, d}}

R es transitiva

R no es antisimétrica, falla en los pares: {{a, b}, {b, a}}

R no es relación de equivalencia

R no es relación de orden
```

Ejercicio 8.8

En el conjunto A={1,2,3,4} se establece la relación binaria
R={{1,1},{2,2},{2,3},{2,4},{3,2},{3,3},{3,4},{4,2},{4,3},{4,4}};
Justificar que es una relación de equivalencia y calcular el conjunto cociente. Comprobar que dicho conjunto de una partición de A.

In[ ]:=
```
A88={1,2,3,4};
R88={{1,1},{2,2},{2,3},{2,4},{3,2},{3,3},{3,4},{4,2},{4,3},{4,4}};
AnalisisRB[A88,R88]
```

```
R es reflexiva

R es simétrica

R es transitiva

R no es antisimétrica, falla en los pares: {{2, 3}, {2, 4}, {3, 2}, {3, 4}, {4, 2}, {4, 3}}

R es una relación de equivalencia

R no es relación de orden
```

In[ ]:=
```
COCIENTE[A88,R88]
```

Out[ ]=
```
{{1}, {2, 3, 4}}
```

Demostración de que A es una partición

In[∘]:=
```
A1={1};
A2={2,3,4};
A1≠{} && A2≠{} (*ningun elemento de la particion es vacio*)
Intersection[A1,A2]=={} (*son disjuntos dos a dos, si hay mas elementos en la particion hay qu
Union[A1,A2]==A88 (*La unión de todos da el total*)
```

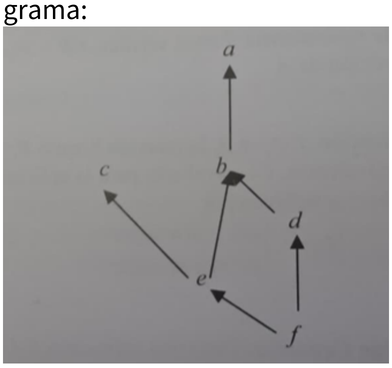
Out[∘]=
True

Out[∘]=
True

Out[∘]=
True

Ejercicio 8.14

Sea X={a,b,c,d,e,f} junto con la ordenación dada por el siguiente diagrama:



Sea Y={b,f,e} y Z={e,b,c,a} subconjunto de X. Se pide:

In[25]:=
```
X814={"a","b","c","d","e","f"};
R814={{"f","f"},{"f","e"},{"f","d"},{"f","c"},{"f","b"},{"f","a"},{"e","e"},{"e","c"},{"e","
{"d","a"},{"b","b"},{"b","a"},{"a","a"},{"c","c"}};
Y814={"b","f","e"};
Z814={"e","b","c","a"};
```

a) Cotas superiores e inferiores de Y y de Z. ¿Existen supremo e ínfimo?

In[31]:=
```
"Cota superior"
COTASSUPERIORES[X814,Y814,R814]
"Supremo"
SUPREMO[X814,Y814,R814]
"Cota inferior"
COTASINFERIORES[X814,Y814,R814]
"Infimo"
INFIMO[X814,Y814,R814]
```

Out[31]=
```
Cota superior
```

Out[32]=
```
{a, b}
```

Out[33]=
```
Supremo
```

Out[34]=
```
{b}
```

Out[35]=
```
Cota inferior
```

Out[36]=
```
{f}
```

Out[37]=
```
Infimo
```

Out[38]=
```
{f}
```

In[39]:=
```
"Cota superior"
COTASSUPERIORES[X814,Z814,R814]
"Supremo"
SUPREMO[X814,Z814,R814]
"Cota inferior"
COTASINFERIORES[X814,Z814,R814]
"Infimo"
INFIMO[X814,Z814,R814]
```

Out[39]=
Cota superior

Out[40]=
{}

Out[41]=
Supremo

Out[42]=
{}

Out[43]=
Cota inferior

Out[44]=
{e, f}

Out[45]=
Infimo

Out[46]=
{e}

b) Máximos y mínimos Y y Z

In[56]:=
```
"Maximo"
MAXIMO[Y814,R814]
"Minimo"
MINIMO[Y814,R814]
```

Out[56]=
Maximo

Out[57]=
{b}

Out[58]=
Minimo

Out[59]=
{f}

In[60]:=
```
"Maximo"
MAXIMO[Z814,R814]
"Minimo"
MINIMO[Z814,R814]
```

Out[60]= Maximo

Out[61]= {}

Out[62]= Minimo

Out[63]= {e}

C) Elementos máximales y minimales de X, Y, Z.

In[ ]:=
```
MAXIMALES[X814,R814]
MINIMALES[X814,R814]
```

Out[ ]= {a, c}

Out[ ]= {f}

In[ ]:=
```
MAXIMALES[Y814,R814]
MINIMALES[Y814,R814]
```

Out[ ]= {b}

Out[ ]= {f}

In[ ]:=
```
MAXIMALES[Z814,R814]
MINIMALES[Z814,R814]
```

Out[ ]= {c, a}

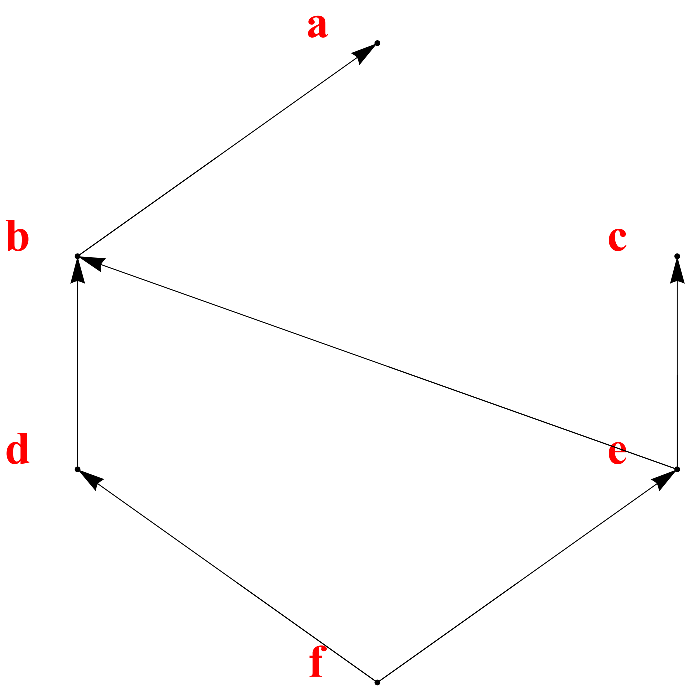
Out[ ]= {e}

b) Representa con Mathematica el diagrama de orden de X, Y y Z

In[ ]:=
```
A=X814;
R=R814;
Clear[Coord];
tabla=Table[0,{i1,Length[A]},{j1,3}];
B=A;
t1=1;
nivel=0;
While[B≠{},minimales={};nivel++;
 Do[minimal=True;
 Do[If[Intersection[{{B⟦m1⟧,B⟦n1⟧}},R]≠{}&&n1≠m1,minimal=False];,{m1,1,Length[B]}];
 If[minimal,AppendTo[minimales,B⟦n1⟧];
 tabla⟦t1⟧={nivel,B⟦n1⟧,0};
 t1++;];,{n1,1,Length[B]}];
 B=Complement[B,minimales];];
R1={};
Do[AppendTo[R1,{A⟦i1⟧,A⟦i1⟧}],{i1,1,Length[A]}];
R=Complement[R,R1];
R1={};
Do[Do[If[Length[Intersection[R,{{R⟦k1,1⟧,A⟦j1⟧},{A⟦j1⟧,R⟦k1,2⟧}}]]==2,R1=Union[R1,{R⟦k1⟧}];]
R=Complement[R,R1];
puntos={};t1=0;
Do[cont=0;
 Do[If[tabla⟦i1,1⟧==j1,cont=cont+1],{i1,1,Length[A]}];
 Do[t1++;
 puntos=Union[puntos,{Text[Style[tabla⟦t1,2⟧,Large,Bold,Red,Background→None,FontFamily→Times]
 tabla⟦t1,3⟧=k1-(cont/2);,{k1,1,cont}],{j1,1,tabla⟦Length[A],1⟧}];
Coord[elem_]:=Do[If[elem==tabla⟦h1,2⟧,Coord[elem]={tabla⟦h1,3⟧,tabla⟦h1,1⟧}],{h1,1,Length[A]
Do[Coord[A⟦i1⟧],{i1,1,Length[A]}];
Do[AppendTo[puntos,Arrow[{Coord[R⟦t1,1⟧],Coord[R⟦t1,2⟧]}]];,{t1,1,Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos],AspectRatio→1,Background→None]
```

Diagrama de orden:

Out[ ]=

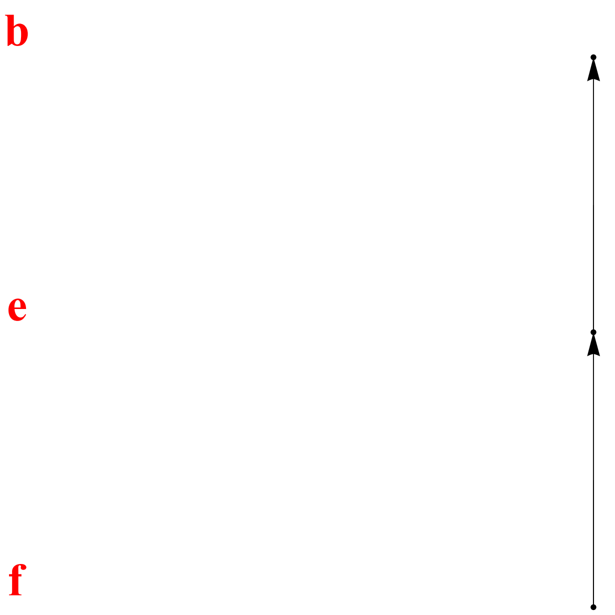


In[∘]:=

```
A=Y814;
R={{"f","f"},{"f","e"},{"f","b"},{"e","e"},{"e","b"},{"b","b"}};
Clear[Coord];
tabla=Table[0,{i1,Length[A]},{j1,3}];
B=A;
t1=1;
nivel=0;
While[B≠{},minimales={};nivel++;
 Do[minimal=True;
 Do[If[Intersection[{{B〚m1〛,B〚n1〛}},R]≠{}&&n1≠m1,minimal=False];,{m1,1,Length[B]}];
 If[minimal,AppendTo[minimales,B〚n1〛];
 tabla〚t1〛={nivel,B〚n1〛,0};
 t1++;];,{n1,1,Length[B]}];
 B=Complement[B,minimales];];
R1={};
Do[AppendTo[R1,{A〚i1〛,A〚i1〛}],{i1,1,Length[A]}];
R=Complement[R,R1];
R1={};
Do[Do[If[Length[Intersection[R,{{R〚k1,1〛,A〚j1〛},{A〚j1〛,R〚k1,2〛}}]]==2,R1=Union[R1,{R〚k1〛}];]
R=Complement[R,R1];
puntos={};t1=0;
Do[cont=0;
 Do[If[tabla〚i1,1〛==j1,cont=cont+1],{i1,1,Length[A]}];
 Do[t1++;
 puntos=Union[puntos,{Text[Style[tabla〚t1,2〛,Large,Bold,Red,Background→None,FontFamily→Times]
 tabla〚t1,3〛=k1-(cont/2);,{k1,1,cont}],{j1,1,tabla〚Length[A],1〛}];
Coord[elem_]:=Do[If[elem==tabla〚h1,2〛,Coord[elem]={tabla〚h1,3〛,tabla〚h1,1〛}],{h1,1,Length[A]
Do[Coord[A〚i1〛],{i1,1,Length[A]}];
Do[AppendTo[puntos,Arrow[{Coord[R〚t1,1〛],Coord[R〚t1,2〛]}]];,{t1,1,Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos],AspectRatio→1,Background→None]
```

Diagrama de orden:

Out[∘]=



In[•]:=
```
A=Z814;
R={{"e","e"},{"e","c"},{"e","b"},{"e","a"},{"b","b"},{"b","a"},{"a","a"},{"c","c"}};;
Clear[Coord];
tabla=Table[0,{i1,Length[A]},{j1,3}];
B=A;
t1=1;
nivel=0;
While[B≠{},minimales={};nivel++;
 Do[minimal=True;
 Do[If[Intersection[{{B⟦m1⟧,B⟦n1⟧}},R]≠{}&&n1≠m1,minimal=False];,{m1,1,Length[B]}];
 If[minimal,AppendTo[minimales,B⟦n1⟧];
 tabla⟦t1⟧={nivel,B⟦n1⟧,0};
 t1++;];,{n1,1,Length[B]}];
 B=Complement[B,minimales];];
R1={};
Do[AppendTo[R1,{A⟦i1⟧,A⟦i1⟧}],{i1,1,Length[A]}];
R=Complement[R,R1];
R1={};
Do[Do[If[Length[Intersection[R,{{R⟦k1,1⟧,A⟦j1⟧},{A⟦j1⟧,R⟦k1,2⟧}}]]==2,R1=Union[R1,{R⟦k1⟧}];]
R=Complement[R,R1];
puntos={};t1=0;
Do[cont=0;
 Do[If[tabla⟦i1,1⟧==j1,cont=cont+1],{i1,1,Length[A]}];
 Do[t1++;
 puntos=Union[puntos,{Text[Style[tabla⟦t1,2⟧,Large,Bold,Red,Background→None,FontFamily→Times]
 tabla⟦t1,3⟧=k1-(cont/2);,{k1,1,cont}],{j1,1,tabla⟦Length[A],1⟧}];
Coord[elem_]:=Do[If[elem==tabla⟦h1,2⟧,Coord[elem]={tabla⟦h1,3⟧,tabla⟦h1,1⟧}],{h1,1,Length[A]
Do[Coord[A⟦i1⟧],{i1,1,Length[A]}];
Do[AppendTo[puntos,Arrow[{Coord[R⟦t1,1⟧],Coord[R⟦t1,2⟧]}]];,{t1,1,Length[R]}];
Print["Diagrama de orden:"]
Show[Graphics[puntos],AspectRatio→1,Background→None]
```

Diagrama de orden:

Out[•]=